Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

COURSE PROJECT

ULTRASONIC DISTANCE MEASUREMENT WITH ALERT SYSTEM

DIGITAL LOGIC DESIGN AND COMPUTER ORGANIZATION (BAITE – 101)

Submitted By:

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Submitted To:

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Components used:

1. Arduino Uno and USB cable
2. HC-SR04 Ultrasonic Sensor
3. OLED Display (0.96”)
4. LEDs (Red, Yellow, Green)
5. Three Resistors (220 Ω)
6. Buzzer
7. Breadboard
8. Jumper Wires (Male to male, male to female)
9. Arduino IDE application and few libraries

Working principle:

This project works using an ultrasonic sensor (HC-SR04), which measures the distance of an object by sending sound waves and calculating the time taken for the echo to return. The Arduino processes this time to calculate the distance and displays it in OLED display then activates the buzzer or LED based on how close the object is.

- Green LED → object far
- Yellow LED → medium distance
- Red LED + buzzer → object too close

The formula used for calculating distance is:

$$Distance = \frac{Time * Speed\ of\ sound}{2}$$

The division by 2 is because the sound wave travels to the object and back.

Arduino Code:

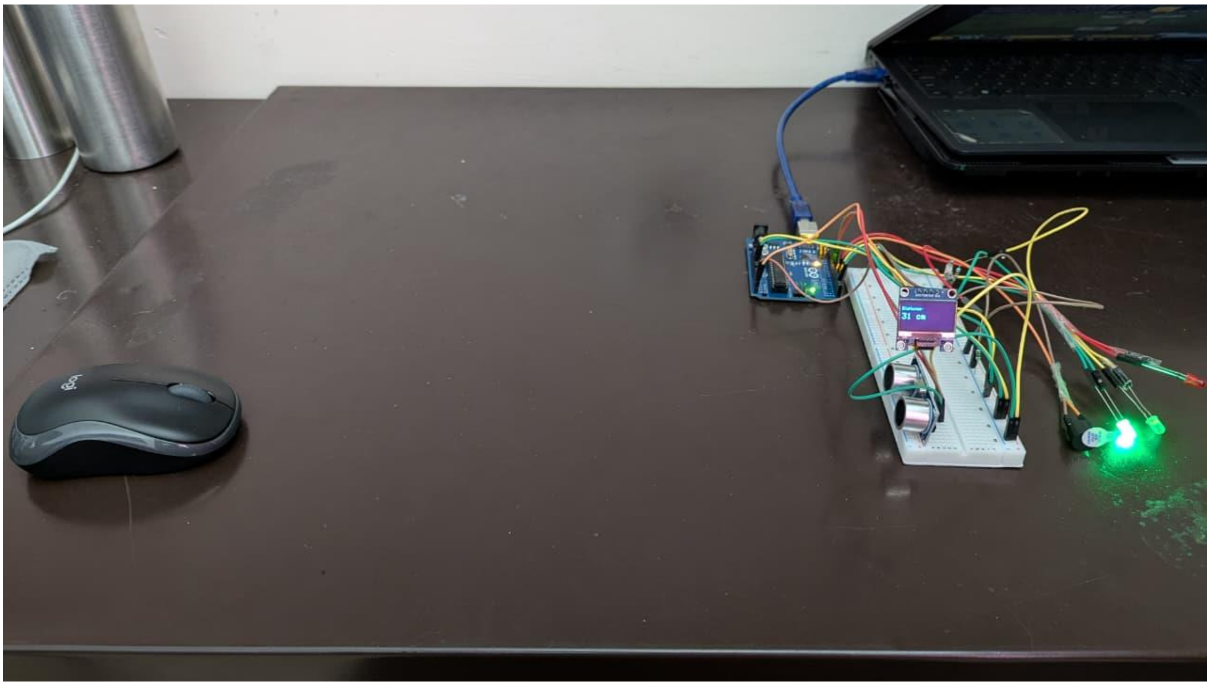
```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -
1);
#define trigPin 9
#define echoPin 10
#define greenLED 2
#define yellowLED 3
#define redLED 4
#define buzzer 5

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(greenLED, OUTPUT);
  pinMode(yellowLED, OUTPUT);
  pinMode(redLED, OUTPUT);
  pinMode(buzzer, OUTPUT);
  Serial.begin(9600);
  // OLED setup
  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println("OLED not found");
    for(;;); // Stop if display not found
  }
```

```
  display.clearDisplay();
  display.setTextSize(1);
  display.setTextColor(SSD1306_WHITE);
  display.setCursor(0,0);
  display.print("Distance Sensor");
  display.display();
  delay(1000);
}
void loop() {
  // Trigger pulse
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Measure echo time
  long duration = pulseIn(echoPin, HIGH);
  int distance = duration * 0.034 / 2;

  // Debug on serial
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  // OLED display
  display.clearDisplay();
  display.setTextSize(1);
  display.setCursor(0, 10);
```

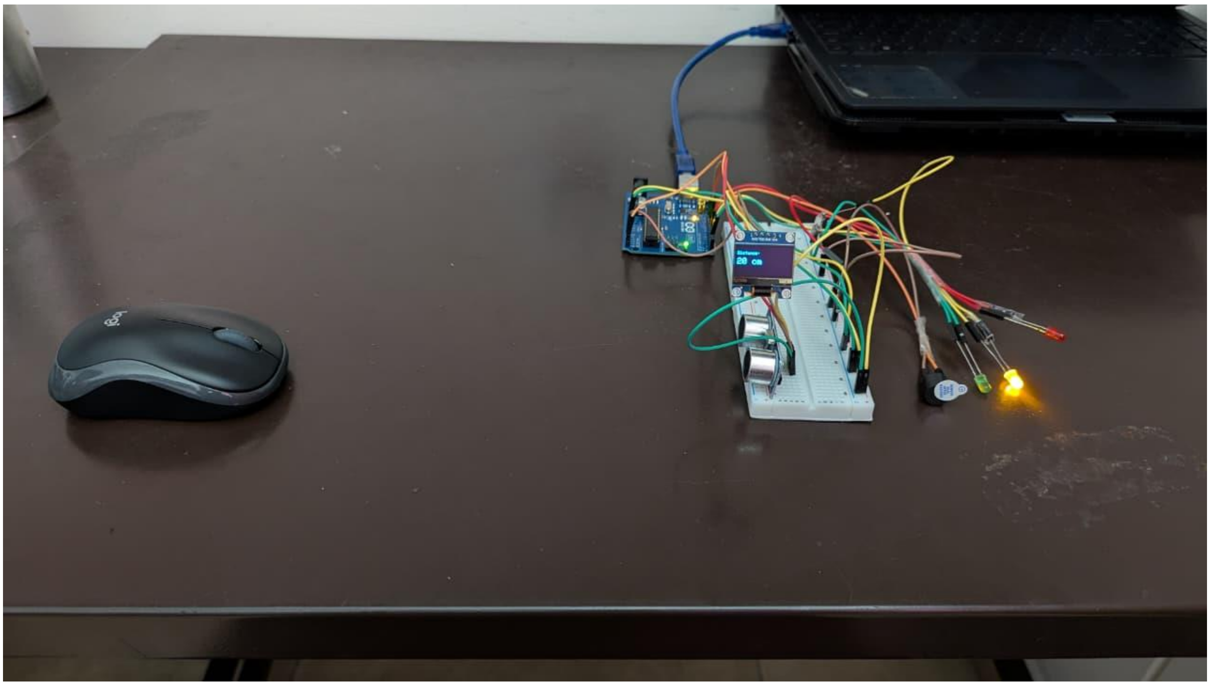
```
  display.print("Distance:");
  display.setTextSize(2);
  display.setCursor(0, 25);
  display.print(distance);
  display.print(" cm");
  display.display();

  // LED & buzzer logic
  if (distance > 30) {
    digitalWrite(greenLED, HIGH);
    digitalWrite(yellowLED, LOW);
    digitalWrite(redLED, LOW);
    digitalWrite(buzzer, LOW);
  }
  else if (distance > 15 && distance <= 30) {
    digitalWrite(greenLED, LOW);
    digitalWrite(yellowLED, HIGH);
    digitalWrite(redLED, LOW);
    digitalWrite(buzzer, LOW);
  }
  else if (distance <= 15) {
    digitalWrite(greenLED, LOW);
    digitalWrite(yellowLED, LOW);
    digitalWrite(redLED, HIGH);
    digitalWrite(buzzer, HIGH);
  }

  delay(300);
}
```

OBSERVATIONS:

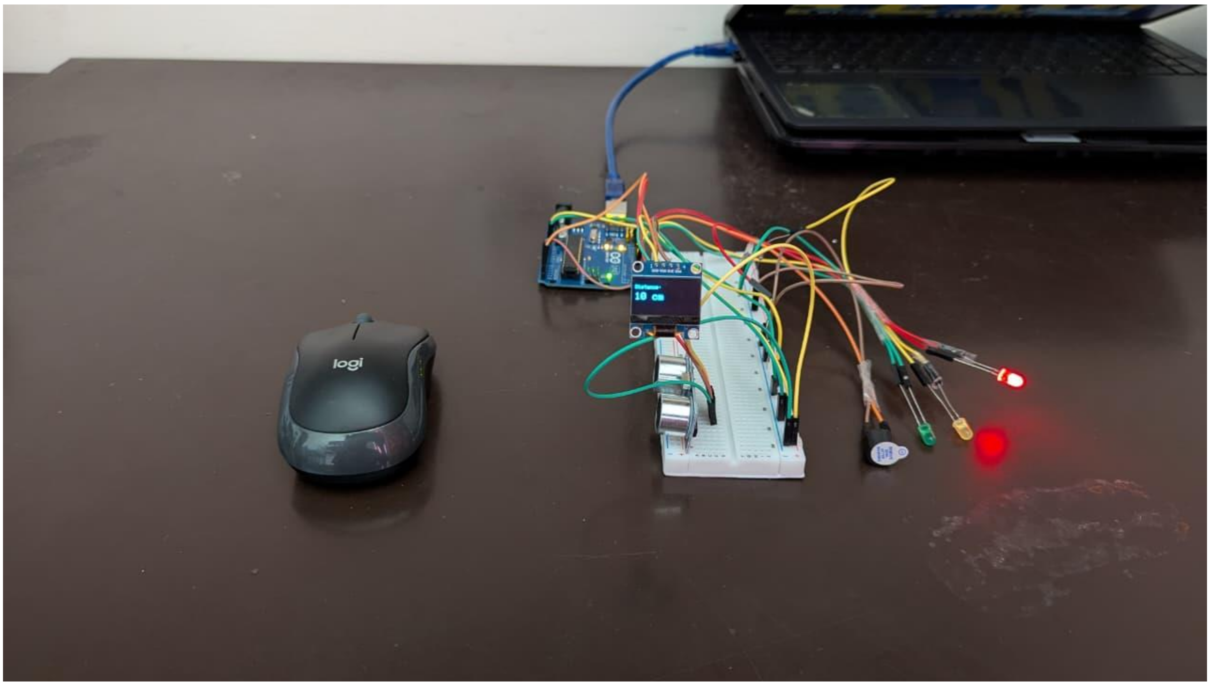
1. If distance > 30 cm, Green LED glows.



2. If distance >15 cm and distance ≤ 30 cm, Yellow LED glows.



3. If distance < 15 cm, Red LED glows and Buzzer starts beeping.



Applications:

i) Parking assistance systems in vehicles

ii) Object detection in robotics

iii) Automatic obstacle alert for blind assistance devices

iv) Industrial safety systems to detect proximity